An Algebraic and Geometric Architecture for a Genuinely Predictive Spin-1 Light-Front Model

Physical motivation, mathematical formulation, computational role, and limits of the proposal

DeuteronWigner project

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Purpose. This note explains a proposed mathematical architecture for the next-level model identified in Section 15 of `references/model_construction_note.tex`. The proposal is not that topology supplies missing QCD dynamics. It is that graded Hilbert spaces, representation-theoretic intertwiners, gauge-bundle parallel transport, symplectic phase-space geometry, and convex positive-state geometry can make the physically required relations structural rather than optional. A more limited chain-complex construction is proposed as an auditable bookkeeping device for Fock-sector transitions and double counting.

Abstract

The present DeuteronWigner model is a constrained phenomenological synthesis: it preserves a common correlator architecture and many exact constraints, but important nucleon, gauge-link, tensor, gluon, and non-nucleonic components are still informed by separately fitted or modeled inputs. A genuinely predictive replacement must instead generate many observables from a small set of shared microscopic interactions and states. We propose organizing that calculation as a gauge-covariant graded light-front tensor category. The dynamical core is a regulated light-front Hamiltonian on flavor-, color-, spin-, orbital-, and Fock-graded Hilbert spaces. Symmetry intertwiners define allowed couplings. Gauge links are parallel transports in an $SU(3)$ bundle and form a representation of a path groupoid. GTMDs are common operator matrix elements whose TMD, GPD, form-factor, current, and OAM reductions must form commuting diagrams. Transverse phase space is treated symplectically, parton–target correlators inhabit a convex positive cone, and nucleon-to-deuteron composition is represented, where applicable, by completely positive amplitude maps. A filtered family of regulated theories defines convergence and renormalization tests. We describe why each layer is physically justified, what it prevents, how it could be implemented, and which stronger topological claims should not be made.

Contents

1 The problem this formulation is meant to solve

The desired advance is a change in the source of prediction. In the present model, many ingredients are physically informed and assembled under a common correlator contract,

$$\{\text{PDF/TMD fits, wave functions, response models, phase models}\} \longrightarrow W_D \longrightarrow \{F_i^D\}. \tag{1}$$

This is scientifically useful, but correlations among different F_i^D are only as fundamental as the assumptions used in the assembly. The next-level direction should instead be

$$\{H_{\rm LF}, \mathcal{R}, \theta\} \longrightarrow \{|\Psi_N\rangle, |\Psi_D\rangle, U_\gamma\} \longrightarrow W_D \longrightarrow \{F_i^D, \mathcal{O}_{\rm process}\}, \tag{2}$$

where $H_{\rm LF}$ is a regulated and renormalized microscopic Hamiltonian, $\mathcal{R}$ denotes the regulator and matching prescription, and θ is a comparatively small set of shared interaction parameters.

The mathematical problem is compositional. A state carries simultaneously Fock sector, flavor, color, parton helicity, target helicity, OAM, gauge-link, and nuclear-component information. Every contraction must respect exact selection rules. Every projected TMD must remain traceable to the same amplitudes. Every truncation must have a convergence meaning. A flat array of independently parameterized functions is a poor representation of this structure.

1.1 Design criterion

We will regard a mathematical structure as useful only if it does at least one of the following:

1. makes an exact symmetry or admissibility condition automatic;
2. prevents an inconsistent or double-counted composition;
3. exposes a controlled approximation and its convergence direction;
4. creates testable relations among observables derived from a common state;
5. reduces redundant parameterization without suppressing a physical degree of freedom.

Abstract terminology that meets none of these criteria should not enter the production model.

2 Graded light-front Hilbert space

2.1 Sector and internal grading

The state space should be a direct sum

$$\mathcal{H}_{\mathrm{LF}} = \bigoplus_{\nu\in\mathcal{F}} \mathcal{H}_\nu, \qquad \nu = (n_q, n_{\bar{q}}, n_g; B, Q, S, \ldots), \tag{3}$$

over a set $\mathcal{F}$ of retained Fock sectors. Each sector has the form

$$\mathcal{H}_\nu = L^2(\mathcal{M}_\nu, \mathrm{d}\mu_\nu) \otimes \mathcal{V}_{\text{flavor}} \otimes \mathcal{V}_{\text{color}} \otimes \mathcal{V}_{\text{spin}} \otimes \mathcal{V}_{\text{OAM}}, \tag{4}$$

where

$$\mathcal{M}_\nu = \left\{ (x_i, \boldsymbol{k}_{iT}) \,\middle|\, x_i > 0, \quad \sum_i x_i = 1, \quad \sum_i \boldsymbol{k}_{iT} = 0 \right\}. \tag{5}$$

The measure contains the conventional light-front phase-space factors and the constraint delta functions.

This formulation gives support and momentum conservation a geometric location: amplitudes are defined on the momentum simplex and transverse constraint surface. Flavor, color, spin, and OAM are not metadata but representation factors acted on by explicit operators.

2.2 Block Hamiltonian and Fock transitions

The mass eigenproblem is

$$H_{\mathrm{LF}}|\Psi_h(P,\Lambda)\rangle = M_h^2|\Psi_h(P,\Lambda)\rangle, \qquad H_{\mathrm{LF}} = P^+P^- - \boldsymbol{P}_T^2. \tag{6}$$

With sector projectors P_ν, define

$$H_{\nu\mu} = P_\nu H_{\mathrm{LF}} P_\mu. \tag{7}$$

The diagonal blocks contain kinetic and sector-preserving interactions; off-diagonal blocks generate emissions, absorption, and pair creation:

$$H_{\mathrm{LF}} \sim \begin{pmatrix} H_{qqq} & V_{qqq,qqqg} & V_{qqq,qqqq\bar{q}} & \cdots \\ V^{\dagger}_{qqq,qqqg} & H_{qqqg} & V_{qqqg,qqqq\bar{q}} & \cdots \\ V^{\dagger}_{qqq,qqqq\bar{q}} & V^{\dagger}_{qqqg,qqqq\bar{q}} & H_{qqqq\bar{q}} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{8}$$

This is the natural setting for basis light-front quantization (BLFQ), Hamiltonian effective field theory, or a compatible continuum bound-state solution mapped to light-front amplitudes [1].

2.3 Why the grading matters

The grading provides explicit answers to questions that are otherwise hidden:

- Which sea distribution is generated by which pair-creation sector?
- Which gluon TMD receives support from $qqqg$ rather than an assigned gluon profile?
- Which spin–orbit observable requires interference between which L_z sectors?
- Which counterterm compensates the integration-out of a sector?
- Does an apparent prediction stabilize as sectors are enlarged?

It also prevents a nucleon gluon input and an explicit $qqqg$ sector from being included without an explicit matching or subtraction rule.

3 Representation theory as the coupling grammar

3.1 Symmetry representations

At a regulated stage, the relevant exact or controlled symmetry action is organized by

$$G = SU(3)_{\mathrm{color}} \times SU(2)_{\mathrm{isospin}} \times SU(2)_{\mathrm{spin}} \times SO(2)_{L_z}, \tag{9}$$

supplemented by discrete parity, time-reversal, and charge-conjugation maps. The $SU(2)_{\mathrm{spin}}$ factor must be understood with the appropriate light-front spin and Melosh/Wigner rotations; full Poincare covariance is a continuum-limit requirement rather than something guaranteed by writing $SU(2)$.

An interaction tensor is an intertwiner

$$C : \mathcal{V}_{R_1} \otimes \cdots \otimes \mathcal{V}_{R_n} \longrightarrow \mathcal{V}_{R_{\mathrm{out}}}, \qquad C\, D_{\mathrm{in}}(g) = D_{\mathrm{out}}(g)\, C. \tag{10}$$

Only invariant or correctly covariant tensors are parameterized. Selection rules are therefore restrictions on the space of maps, not penalties added after fitting.

3.2 Spin-1 irreducible density structure

For a spin-1 target,

$$1 \otimes 1^* = 0 \oplus 1 \oplus 2. \tag{11}$$

Accordingly, the density operator decomposes into scalar, vector, and quadrupole pieces,

$$\rho_D = \rho_D^{(0)} + \sum_{m=-1}^{1} S_m\, T_m^{(1)} + \sum_{m=-2}^{2} Q_m\, T_m^{(2)}. \tag{12}$$

The familiar U, L, T, LL, LT, TT labels are basis-dependent real combinations of these irreducible components. Working first in the spherical-tensor basis has concrete benefits:

- rotations are implemented by Wigner D-matrices;
- vector and tensor responses cannot be silently conflated;
- pure-S, S–D, and additional OAM limits have explicit Clebsch–Gordan selection rules;
- the Cartesian TMD basis is produced by a fixed invertible map.

3.3 Flavor and controlled isospin

Proton and neutron states should inhabit an isospin multiplet, but physical operators may include controlled breaking:

$$H_{\mathrm{LF}} = H_0 + \delta H_{m_d-m_u} + \delta H_{\mathrm{QED}} + \delta H_{\mathrm{nuclear}}. \tag{13}$$

Exact isospin is recovered when the last three terms are disabled. It is not implemented by assigning identical u and d functions. The representation structure therefore permits both exact interchange tests and realistic charge-symmetry breaking.

3.4 Color invariants and gluon T-odd sectors

The two three-adjoint invariant tensors

$$f^{abc}, \qquad d^{abc} \tag{14}$$

define inequivalent color contractions. They should be different intertwiner channels in the operator graph. This prevents a common scalar "gluon T-odd amplitude" from being copied into both sectors and allows a hard process to select its own linear combination.

4 A symmetry-preserving tensor-network realization

4.1 What the network represents

A tensor network is proposed as a sparse computational representation of the amplitude and operator contractions, not as an assumption that QCD is one-dimensional or has low entanglement everywhere. Edges carry

$$e = (\nu, R_c, R_I, j, \lambda, L_z, \alpha), \tag{15}$$

and nodes are physical maps: Hamiltonian vertices, spin couplers, wave-function coefficients, current insertions, Wilson-line vertices, or nuclear Fock-sector couplers.

For example, a schematic deuteron amplitude is

$$|\Psi_D^{\Lambda}\rangle = \mathcal{C}_D^{\Lambda}\big[|\Psi_p\rangle \otimes |\Psi_n\rangle \oplus |\Psi_{NN\pi}\rangle \oplus |\Psi_{\Delta\Delta}\rangle \oplus |\Psi_{6q}\rangle + \cdots\big]. \tag{16}$$

The NN coupling contains maps

$$C_{D,L} : \left(\tfrac{1}{2} \otimes \tfrac{1}{2}\right)_{S=1} \otimes L \longrightarrow J = 1, \qquad L = 0, 2, \ldots. \tag{17}$$

4.2 What it buys us

The network makes shared microscopic origin inspectable. A parameter lives on one Hamiltonian vertex or state tensor and propagates to every observable that uses it. It cannot become an independent coefficient for each TMD. Automatic differentiation through contractions can produce correlated sensitivities

$$J_{ia} = \frac{\partial \mathcal{O}_i}{\partial \theta_a}, \qquad \mathrm{Cov}(\mathcal{O}) \simeq J\ \mathrm{Cov}(\theta)\, J^T, \tag{18}$$

while retaining the exact symmetry blocks.

4.3 What it does not buy us

A tensor network does not make the interaction correct. Aggressive low-rank truncation may discard precisely the OAM, color, or long-range correlations of interest. Bond-dimension convergence must therefore be treated on the same footing as basis and Fock-sector convergence. The network is a representation and contraction strategy, not a phenomenological replacement for H_{LF}.

5 Gauge links as bundle parallel transport

5.1 Connection and holonomy

Let $P(M, SU(3))$ be the color principal bundle over the relevant spacetime domain and

$$A = A^a_\mu t^a \, \mathrm{d}x^\mu \tag{19}$$

its connection. The gauge link along an oriented path $\gamma : a \to b$ is

$$U_\gamma[b,a] = \mathcal{P} \exp\left[-ig \int_\gamma A\right]. \tag{20}$$

Parallel transport obeys the exact algebra

$$U_{\gamma_2 \circ \gamma_1} = U_{\gamma_2} U_{\gamma_1}, \qquad U_{\gamma^{-1}} = U_\gamma^\dagger. \tag{21}$$

Quarks use the fundamental representation and gluons the adjoint representation. Staple direction, transverse closure, rapidity regulator, and representation are properties of the operator itself.

5.2 The path groupoid

The admissible gauge-link paths form a groupoid:

$$\mathrm{Obj}(\mathrm{Path}) = \{\text{operator endpoints}\}, \tag{22}$$
$$\mathrm{Hom}_{\mathrm{Path}}(a,b) = \{\gamma : a \to b\}, \tag{23}$$
$$\gamma^{-1} : b \to a. \tag{24}$$

The Wilson-line assignment is a representation

$$\mathcal{U}_R : \mathrm{Path} \longrightarrow \mathrm{End}(\mathcal{V}_R), \qquad \gamma \longmapsto U_\gamma^{(R)}. \tag{25}$$

Future- and past-pointing staples are distinct morphisms. Time reversal maps their classes into one another, so the familiar T-even/T-odd relations are implemented as transformations of operators:

$$F_{\text{even}}^{[\gamma+]} = F_{\text{even}}^{[\gamma-]}, \tag{26}$$

$$F_{\text{odd}}^{[\gamma+]} = -F_{\text{odd}}^{[\gamma-]}, \tag{27}$$

subject to the relevant conjugations and conventions.

5.3 Where the dynamics enters

The curvature

$$F = \mathrm{d}A - igA \wedge A \tag{28}$$

controls local path dependence. Expanding the holonomy,

$$U_\gamma = 1 - ig \int_\gamma A - g^2 \int_\gamma A\,A + \cdots , \tag{29}$$

shows that absorptive phases and naive-T-odd functions must arise from intermediate states, pole prescriptions, and rescattering dynamics. The path groupoid organizes which phase is allowed; it does not determine the phase numerically. That requires the Hamiltonian gauge field and the factorization-appropriate eikonal limit.

5.4 Why this is geometric but not topological QCD

For a flat connection, holonomy may depend only on homotopy class. QCD generically has $F_{\mu\nu} \neq 0$, so the Wilson line depends on the detailed path geometry. It is therefore incorrect to call a Sivers or Boer–Mulders function a topological invariant. Topology can classify link connectivity and closed cycles; the non-Abelian connection supplies their physical values. This distinction is essential if the construction is to remain falsifiable.

6 GTMDs as common morphisms and commuting reductions

6.1 Operator matrix element

For an operator label Γ and link path γ, define

$$\mathcal{W}_{\Gamma,\gamma} : \mathcal{H}_h \longrightarrow \mathcal{H}_h. \tag{30}$$

Its off-forward matrix element is

$$W_{\Lambda'\Lambda}^{[\Gamma,\gamma]}(x, \boldsymbol{k}_T, \boldsymbol{\Delta}_T) = \langle P', \Lambda' | \mathcal{W}_{\Gamma,\gamma}(x, \boldsymbol{k}_T, \boldsymbol{\Delta}_T) | P, \Lambda \rangle. \tag{31}$$

A wave-function calculation evaluates equation (31) as overlaps of the same Fock amplitudes, including diagonal and, where required, off-diagonal sector terms. GTMDs supply a natural common ancestor for TMD and GPD limits [2].

6.2 Reduction maps

Define schematic linear maps

$$\begin{aligned} \mathrm{T}[W] &= W|_{\boldsymbol{\Delta}_T=0}, & \mathrm{G}[W] &= \int \mathrm{d}^2\boldsymbol{k}_T\, W, \\ \mathrm{F}[W] &= \int \mathrm{d}x\, \mathrm{d}^2\boldsymbol{k}_T\, W, & \mathrm{Q}_\Gamma[W] &= \text{local-current reduction.} \end{aligned} \tag{32}$$

The predictive architecture requires compatible reductions to commute:

$$\mathrm{F} \circ \mathrm{T}[W] = \mathrm{Q}_\Gamma[W] \tag{33}$$

where the regulator, matching scheme, and operator definitions make the identity applicable. Similarly, nuclear composition N and projection P_i should satisfy

$$\mathrm{P}_i \circ \mathrm{N}[W_N] \stackrel{?}{=} \mathrm{N}_i \circ \mathrm{P}[W_N]. \tag{34}$$

Failure of equation (34) is not automatically an error: it may show that projection-dependent many-body currents or coherent terms are missing. The commuting diagram is therefore both an architecture and a diagnostic.

6.3 Why this is more predictive

An independently fitted f_1, g_1, h_1, Sivers function, and pretzelosity need not satisfy any common off-forward, current, or OAM relation. If all are reductions of equation (31), their correlations follow from the state and operator. A failure against a withheld observable then challenges the shared amplitudes rather than one isolated curve.

7 Symplectic geometry of transverse phase space

7.1 Phase space and moment map

The transverse Wigner variables inhabit

$$T^*\mathbb{R}^2 = \{(\boldsymbol{b}_T, \boldsymbol{k}_T)\}, \tag{35}$$

with canonical symplectic form

$$\omega = \mathrm{d}b_x \wedge \mathrm{d}k_x + \mathrm{d}b_y \wedge \mathrm{d}k_y. \tag{36}$$

The $SO(2)$ rotation action has moment map

$$\mu_{L_z}(\boldsymbol{b}_T, \boldsymbol{k}_T) = b_x k_y - b_y k_x. \tag{37}$$

Thus an OAM expectation is a phase-space moment

$$\langle L_z \rangle = \int \mathrm{d}x\, \mathrm{d}^2\boldsymbol{b}_T\, \mathrm{d}^2\boldsymbol{k}_T\, \mu_{L_z}(\boldsymbol{b}_T, \boldsymbol{k}_T)\, \rho_W(x, \boldsymbol{b}_T, \boldsymbol{k}_T; \gamma), \tag{38}$$

with an operator definition that retains its Wilson-line dependence [3].

7.2 Transverse rank as representation weight

Functions $e^{im\phi}$ carry irreducible $SO(2)$ weight m. Transverse-rank tensors are therefore organized by angular harmonics rather than by visually assigned powers of k_T. A matrix element obeys

$$\Delta L_z + \Delta\lambda_{\text{parton}} + \Delta\Lambda_{\text{target}} = m_{\text{operator}}. \tag{39}$$

Equation (39) identifies which partial-wave interferences can contribute to worm gears, pretzelosity, tensor TMDs, and linearly polarized gluon distributions. It also supplies controlled zero tests: a required interference vanishes when either participating amplitude is removed.

7.3 Caution about Wigner functions

The Wigner distribution is a quasiprobability and need not be pointwise positive. Symplectic geometry organizes its transformations and moments; it must not be confused with the positive density-matrix cone discussed next.

8 Convex geometry and positivity by construction

8.1 The correlator cone

At fixed kinematics, the combined parton–target helicity correlator must be Hermitian and positive semidefinite in the applicable forward probability interpretation:

$$\Phi(x, \boldsymbol{k}_T) \in \mathrm{CP}_n \equiv \{X = X^\dagger \mid X \succeq 0\}. \tag{40}$$

Named TMDs are linear coordinates on the image of this cone under a projection map. Their allowed region is therefore a convex spectrahedral set, not a Cartesian box of independent error bars.

The most physical factorization is

$$\Phi_{\alpha\beta} = \sum_X A_{\alpha X} A^*_{\beta X}, \tag{41}$$

where X labels intermediate or spectator states. Numerically one may also use a Cholesky-like form

$$\Phi = LL^\dagger. \tag{42}$$

Hermiticity and positivity then hold for every parameter value. Principal minors yield Soffer-like and spin-1 tensor bounds. This density-matrix view is already the basis of known spin-1 gluon TMD positivity constraints [4].

8.2 Why clipping is unacceptable

If independently generated TMDs are assembled and negative eigenvalues are clipped, the model changes in a nonlinear and untraceable way. In equation (41), a positivity failure instead means that the claimed amplitudes or projection map are inconsistent. Uncertainty samples remain inside the physical cone without erasing correlations among different functions.

8.3 Information geometry

For normalized positive states, distinguishability may be quantified by the quantum Fisher metric or relative entropy. This has a concrete possible use in experiment design: identify observables that separate two microscopic mechanisms rather than merely enlarging an already constrained direction. It should be introduced only after the physical density operator and likelihood are fixed.

9 Nucleon-to-deuteron composition as an amplitude map

9.1 Completely positive impulse composition

For an incoherent retained subsystem, the nucleon-to-deuteron map can be written

$$\mathcal{E}_D(\rho_N) = \sum_\alpha K_\alpha \rho_N K_\alpha^\dagger, \tag{43}$$

where α includes spectator helicity, partial wave, recoil, and retained nuclear channel. If the map is trace preserving,

$$\sum_\alpha K_\alpha^\dagger K_\alpha = I. \tag{44}$$

If probability flows to $NN\pi$, $\Delta\Delta$, or other explicit sectors, the corresponding maps must be added before imposing total normalization.

The value of equation (43) is that positivity is inherited from the nucleon correlator. Tensor polarization is generated by the angular and spin structure of the K_α, rather than by multiplying an unpolarized answer by an arbitrary tensor factor.

9.2 Limits of the quantum-channel picture

Coherent shadowing, phases between Fock sectors, and many-body currents must be computed at the amplitude level before an environmental trace. An apparently non-completely-positive reduced approximation may indicate that a coherent degree of freedom was traced out inconsistently. We should not force every nuclear mechanism into an incoherent Kraus map. Instead, the full amplitude is primary and the channel representation is used where its assumptions are satisfied.

10 Chain complexes for composition and double counting

10.1 A deliberately limited topological layer

Let C_n be the vector space generated by retained configurations with n resolved transitions relative to a reference sector. A boundary map

$$\partial_n : C_n \longrightarrow C_{n-1}, \qquad \partial_{n-1}\partial_n = 0, \tag{45}$$

records the endpoints and signs of sector-transition sequences. Examples include

$$NN \leftrightarrow NN\pi, \qquad qqq \leftrightarrow qqqg, \qquad qqq \leftrightarrow qqqq\bar{q}.$$

This layer can support:

- automated detection of the same pion dressing represented both inside a nucleon input and as an explicit $NN\pi$ sector;
- association of effective operators and counterterms with the sectors that were integrated out;
- classification of closed transition cycles requiring interference or unitarity checks;
- provenance paths from a final observable back to its amplitude components.

The homology

$$H_n = \ker \partial_n / \operatorname{im} \partial_{n+1} \tag{46}$$

would classify independent cycles not reducible to already represented higher transitions. This is initially an audit construction, not a claim that physical amplitudes depend only on a homology class.

10.2 Why category language is also useful

Objects can represent typed state/operator spaces; morphisms represent Hamiltonian evolution, Wilson transport, projection, matching, evolution, or nuclear composition. Composition is allowed only when domains, codomains, schemes, link classes, and quantum numbers match. This provides a mathematically precise version of the project's fail-closed typed interfaces.

We do not need sophisticated category theory to implement this. Sparse block tensors plus checked domain/codomain metadata are sufficient. The categorical viewpoint is valuable because it states what must compose and which diagrams must commute.

11 Regulator, renormalization, and convergence as a filtration

11.1 Nested regulated spaces

Define a family of calculations

$$\mathcal{H}_1 \hookrightarrow \mathcal{H}_2 \hookrightarrow \cdots, \qquad \mathcal{H}_n = \mathcal{H}(\Lambda_n, N_{\max,n}, \mathcal{F}_n), \tag{47}$$

that enlarges ultraviolet resolution, basis size, and Fock content. Each level has

$$H^{(n)} = H_0^{(n)} + \sum_a c_a^{(n)} O_a^{(n)}. \tag{48}$$

Matching maps

$$R_{n+1\to n} : \mathcal{A}_{n+1} \longrightarrow \mathcal{A}_n \tag{49}$$

relate effective operators across truncations. For calibration observables,

$$R_{n+1\to n}[\mathcal{O}^{(n+1)}] = \mathcal{O}^{(n)} + \delta_n, \qquad \delta_n \to 0 \tag{50}$$

within an estimated truncation law.

11.2 Acceptance meaning

A smooth result at one $N_{\max}$ is not a prediction. A predictive claim requires stability under:

- basis enlargement;
- regulator variation with counterterm refitting;

- addition of the next physically required Fock sector;
- increased tensor-network bond or factorization rank;
- quadrature and grid refinement.

The correlated spread across these directions is a model-discrepancy component, not interchangeable with posterior parameter uncertainty.

12 Inference and falsifiability

12.1 Shared microscopic parameters

Let θ contain masses, couplings, regulator-dependent counterterms, and a restricted number of effective interactions. The posterior is

$$p(\theta, \delta_{\mathrm{EFT}} \mid D_{\mathrm{cal}}) \propto p(D_{\mathrm{cal}} \mid \theta, \delta_{\mathrm{EFT}}) p(\theta) p(\delta_{\mathrm{EFT}}). \tag{51}$$

The calibration data may include masses, elastic currents, selected nucleon PDFs/charges, deuteron static properties, and a limited set of process observables. It must not include a free normalization and transverse width for every TMD.

The predictive distribution for withheld data is

$$p(D_{\mathrm{hold}} \mid D_{\mathrm{cal}}) = \int \mathrm{d}\theta \, \mathrm{d}\delta_{\mathrm{EFT}} \, p(D_{\mathrm{hold}} \mid \theta, \delta_{\mathrm{EFT}}) p(\theta, \delta_{\mathrm{EFT}} \mid D_{\mathrm{cal}}). \tag{52}$$

A failed withheld family must revise the state, interaction, current, or truncation model. Adding a coefficient directly to the failed TMD would return us to the patchwork construction.

12.2 The prediction hierarchy

The strongest test is cross-channel:

1. calibrate a common Hamiltonian using spectroscopy, currents, and a restricted set of partonic data;
2. predict GTMD marginals not used in calibration;
3. predict link-reversal or color-channel relations;
4. predict tensor-polarized and gluon observables;
5. propagate without retuning into deuteron observables.

The framework is worthwhile only if it makes such tests possible.

13 Software realization

13.1 Core objects

Object	Responsibility
`SectorSpace`	Momentum manifold, basis, Fock grade, representations, regulator, and normalization.
`Intertwiner`	Sparse invariant tensor with explicit domain, codomain, multiplicity, and symmetry tests.
`LFHamiltonian`	Hermitian block operator, counterterms, sector-changing vertices, and eigensolver adapter.
`FockState`	Normalized amplitudes, phase convention, member identity, and truncation metadata.
`PathGroupoid` and `WilsonTransport`	Admissible link paths, inverse/composition laws, color representation, rapidity scheme, and dynamical holonomy evaluation.
`GTMDOperator`	Off-forward operator insertion and common Fock-overlap evaluator.
`ReductionMap`	Typed TMD, GPD, current, form-factor, and OAM reductions with commuting diagram tests.
`NuclearAmplitudeMap`	Spin-1 Fock composition, coherent mechanisms, currents, and controlled partial trace.
`PositiveCorrelator`	Amplitude or $LL^\dagger$ representation and irreducible TMD coordinates.
`TruncationTower`	Regulator/basis/Fock filtration, matching maps, convergence fits, and model-discrepancy estimates.
`ProvenanceComplex`	Transition graph, no-double-counting audit, source/replacement information, and observable ancestry.

13.2 Required property-based tests

1. Intertwiners commute with every represented symmetry generator.
2. $H_{\mathrm{LF}} = H_{\mathrm{LF}}^\dagger$, and its Poincare commutator defects decrease along the truncation tower.
3. Wilson transport respects path concatenation and inversion.
4. Time reversal maps future and past link operators with the correct T-even/T-odd behavior.
5. GTMD reduction diagrams commute within regulator and numerical tolerances.
6. Density correlators remain positive without eigenvalue clipping.
7. Spin-1 rotations reproduce the irreducible $0 \oplus 1 \oplus 2$ transformation law.
8. Nuclear maps close total probability and momentum only after all declared Fock sectors are included.
9. Disabling either amplitude in an OAM interference removes the corresponding harmonic.
10. Sector-transition provenance detects deliberately duplicated pion, gluon, or sea mechanisms.

13.3 Role of PennyLane

PennyLane could prototype small symmetry-reduced Hamiltonian blocks or state-composition circuits only after specifying the physical basis, Hamiltonian, observables, and encoding. A circuit should be accepted only if it reproduces exact diagonalization in the same truncated Hilbert space and offers a concrete advantage for variational eigensolving, differentiation, or correlation diagnostics. A generic entangling circuit with no map to Fock amplitudes would add parameters without physics and should be rejected.

14 Staged research program

14.1 Stage A: algebraic skeleton with analytic benchmarks

1. Implement irreducible color, isospin, spin-1, and $SO(2)$ OAM labels.
2. Encode the existing model's correlators as positive block tensors without changing their central physics.
3. Implement path-groupoid link types and exact transformation laws.
4. Establish commuting reductions on analytic two- and three-body light-front models.

This stage tests the architecture without claiming new dynamics.

14.2 Stage B: microscopic nucleon pilot

Choose one controlled Hamiltonian route and solve at least

$$|qqq\rangle \oplus |qqqg\rangle \oplus |qqqq\bar{q}\rangle$$

at multiple truncations. Compute a nonzero-$\boldsymbol{\Delta}_T$ quark/gluon GTMD subset, its TMD/GPD/current reductions, and OAM moments. Compare with known nucleon charges, PDFs, form factors, and lattice moments. The pilot is successful only if more than one observable family is predicted from shared parameters.

14.3 Stage C: dynamical gauge-link pilot

Add an eikonal Wilson-line interaction derived from the same gauge sector. Test:

- vanishing of naive-T-odd functions at zero rescattering;
- exact future/past reversal;
- flavor and operator differences from shared intermediate states;
- separate gluon f- and d-type contractions;
- stability against regulator and Fock enlargement.

14.4 Stage D: spin-1 microscopic composition

Construct the deuteron from a consistent NN interaction and current, then add $NN\pi$ and any retained non-nucleonic sectors with normalization and subtraction rules. Reproduce binding, static moments, elastic form factors, and controlled b_1 limits before predicting the full tensor TMD sector.

14.5 Stage E: inference, evolution, and withheld validation

Match to a declared TMD scheme, evolve all ranks and coupled partonic sectors, calibrate shared parameters, and reserve at least one observable family for posterior predictive validation. This stage is the first point at which the phrase "genuinely predictive next-level model" could be audited.

15 Risks and explicit nonclaims

Risk	Failure mode	Required safeguard
Decorative topology	Calling link classes or TMDs topological invariants without a dynamical derivation.	Restrict topology to path/transition organization; retain connection and Hamiltonian dependence.
Symmetry overconstraint	Exact isospin or truncated rotational symmetry erases physical breaking.	Implement breaking operators and test the symmetric limit separately.
Tensor-network bias	Low bond dimension suppresses long-range or OAM correlations.	Publish bond-rank convergence alongside basis/Fock convergence.
Positivity misuse	Demanding pointwise positivity of a Wigner quasidistribution.	Apply convex positivity to the appropriate forward helicity correlator, not to all phase-space representations.
Channel misuse	Forcing coherent shadowing into an incoherent completely positive map.	Retain amplitude phases and trace only after the coherent calculation.
Renormalization gap	Treating a finite Hamiltonian matrix as QCD without counterterms or matching.	Specify regulator, counterterms, matching observables, and truncation flow.
Parameter migration	Adding one latent coefficient per TMD inside sophisticated notation.	Attach parameters only to microscopic vertices, currents, or controlled discrepancy operators.

16 Recommended decision

The recommended foundation is a *gauge-covariant graded light-front tensor architecture with convex-state geometry*. Its indispensable elements are:

1. the graded regulated light-front Hilbert space and Hamiltonian;
2. representation-theoretic intertwiners for color, flavor, spin, OAM, and spin-1 polarization;
3. common GTMD operators and commuting observable reductions;
4. Wilson-line parallel transport represented on an admissible path groupoid;
5. symplectic organization of transverse phase space and OAM;
6. positive-semidefinite amplitude geometry for forward correlators;

7. a filtered truncation tower with matching and convergence.

The tensor-network implementation is strongly recommended as the sparse computational realization. The chain-complex layer is recommended initially as a provenance and no-double-counting audit. More ambitious topological constructions should be deferred until a concrete physical observable or computational obstruction requires them.

The proposal does not generate the interaction Hamiltonian, determine counterterms, or solve QCD by terminology. Its value is stricter and more practical: it gives the microscopic calculation a structure in which flavor, spin, OAM, gauge link, color, nuclear composition, positivity, and uncertainty cannot drift apart. If implemented faithfully, it would convert many of the project's present validation tests into consequences of the representation and would make the remaining failures scientifically diagnostic.

A Compact mathematical dictionary

Physical concept	Mathematical object
Fock sectors	Grading and direct-sum Hilbert space
Momentum support	Simplex/constraint manifold $\mathcal{M}_\nu$
Flavor, color, spin, OAM	Group representations and irreducible labels
Allowed interaction	Equivariant intertwiner
Spin-1 vector/tensor polarization	Rank 1 and 2 spherical tensors
Sparse state contraction	Symmetry-preserving tensor network
Gauge link	Bundle parallel transport/holonomy
Process-dependent staple	Morphism in a path groupoid
GTMD	Gauge-covariant operator morphism/matrix element
TMD/GPD/current/OAM limits	Typed reduction maps
Consistency identity	Commuting diagram
Transverse OAM	Moment map on $T^*\mathbb{R}^2$
Forward positivity	Positive-semidefinite convex cone
Impulse nuclear dressing	Completely positive amplitude map where valid
Sector bookkeeping	Chain complex/provenance graph
Truncation sequence	Filtered Hilbert spaces/projective matching system
Predictive uncertainty	Pushforward of shared posterior plus discrepancy

B Acceptance questions for adopting the architecture

1. Can every tensor index be mapped to a physical degree of freedom or controlled numerical basis label?
2. Does every parameter belong to a Hamiltonian term, state amplitude, current, matching coefficient, or explicit discrepancy operator?
3. Are Wilson paths and color representations part of operator identity?

4. Can at least two different observable reductions be computed from the same nonzero-transfer correlator?
5. Is positivity structural rather than repaired?
6. Can exact isospin, pure-S, zero-rescattering, and zero-additional- Fock-sector limits be recovered?
7. Is every approximation embedded in a convergence sequence?
8. Does the architecture enable a genuinely withheld prediction?

A negative answer does not necessarily reject the entire formulation, but it identifies the next concrete design or physics problem.

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